

Research Statement

SUMMARY

My primary research interests lie in algebraic geometry, commutative algebra, and algebraic combinatorics. Much of my research is motivated in some way by problems from enumerative combinatorics; here, one can think geometrically, prove algebraically, and compute combinatorially.

The graded Möbius algebra, arising in the recent proof of the Top-Heavy Conjecture [8], encodes the combinatorics of a matroid's lattice of flats. Work of LaClair, Mastroeni, McCullough, and Peeva characterized which graded Möbius algebras of graphic matroids are Koszul [23]. In recent work with McCullough, we study the Koszul property for the graded Möbius algebras of matroids associated to projective and affine geometries [17], and with a larger group of collaborators, the homological properties of these algebras [12].

Lefschetz properties are a powerful tool to prove statements about unimodality and log-concavity in enumerative combinatorics (e.g., the resolutions of conjectures of Heron–Rota–Welsh and Mason–Welsh in [2]). In the same spirit as the hard Lefschetz theorem, the weak and strong Lefschetz properties can be defined for an Artinian algebra. In joint work with Schenck, we showed that Artinian algebras with a certain pattern in their Betti tables—an organized way to summarize graded minimal free resolutions—fail to have the weak Lefschetz property [18]. In related work, my collaborators and I explicitly compute the graded Betti numbers of connected sums of Artinian Gorenstein algebras [4].

Parallel to this research, I have also been working with collaborators on describing the combinatorics of the stable Tamari lattice, stemming from work of Haiman [19]. This includes determining the cover relations, height statistics, and sizes of the lower order ideals [9, 10]. The research in this project is meant to provide a combinatorial framework for problems extending from diagonal harmonics, which has already given rise to results such as the shuffle theorem and the generalized delta square conjecture.

NOTATION

Let $R = \mathbb{K}[x_1, \dots, x_n]$ be a polynomial ring over a field $\mathbb{K}$ and M a finitely generated R -module. A *free resolution* of M over R is an exact complex of free R -modules

$$F_\bullet : F_0 \leftarrow F_1 \leftarrow \cdots \leftarrow F_p \leftarrow 0$$

such that $M \cong \operatorname{coker}(F_1 \rightarrow F_0)$. Since the F_i are free, we can consider the maps in the resolution as matrices with entries from R . The resolution $F_\bullet$ is *minimal* if the entries in these maps lie in the maximal ideal $\mathfrak{m} = (x_1, \dots, x_n)$. When M is graded, $F_\bullet$ can be chosen to be a *graded* minimal free resolution of M so that $F_i = \bigoplus_j R(-j)^{\beta_{i,j}}$, where $R(-j)$ is a rank 1 free R -module with generator in degree j .

We use M_i to denote the $\mathbb{K}$ -vector space generated by the homogeneous elements of M of degree i . The Hilbert function $H_M(i)$ is then defined as $H_M(i) = \dim_{\mathbb{K}}(M_i)$, and the Hilbert series is $\operatorname{HS}_M(t) = \sum_{i \geq 0} H_M(i)t^i$. Importantly, the (graded) *Betti numbers* of M consist of the $\beta_{i,j}$; these serve as numerical invariants which capture the complexity of M . The Betti numbers of M can be arranged into the *Betti table* of M , where the entry in column i and row j is $\beta_{i,i+j}$. The value $\beta_i = \sum_j \beta_{i,j}$ is the i -th *total Betti number* of M and can be collected in the *Poincaré series* $P_M^R(t) := \sum_{i \geq 0} \beta_i t^i$.

A matroid is a combinatorial abstraction of the notion of linear independence for vector spaces. Matroids have a rich combinatorial structure (they are simplicial complexes with an “exchange property”), and much work has been put into computing their enumerative statistics. Of note, the Dowling-Wilson Top-Heavy Conjecture asserted that the lattice of flats of a simple matroid is “top heavy” in the sense that if M is a simple matroid of rank r and W_i is the number of flats of rank i , then $W_i \leq W_{r-i}$ for all $0 \leq i \leq \frac{r}{2}$. This conjecture remained open for over 40 years until Huh and Wang resolved it when M is representable [21], and then Braden et al. proved it in full generality, in part, by using the combinatorics of the graded Möbius algebra $\text{GMA}(M)$ (defined in [7]) and Lefschetz-type theorems of the Chow ring $\text{CH}(M)$ and the augmented Chow ring $\text{CH}(M)$ [8].

1.1. Koszulness. A standard graded $\mathbb{K}$ -algebra S is *Koszul* if $\mathbb{K}$ has a linear free resolution over S . Koszul algebras appear in many contexts, for example, as homogeneous coordinate rings of general canonical curves and of Schubert varieties. A standard result shows that if an algebra is Koszul, then it is quadratic, while the converse is false in general. Beyond contraining the graded minimal free resolution of $\mathbb{K}$ over S , the Koszul property imposes the identity $P_{\mathbb{K}}^S(t) \cdot H_S(-t) = 1$.

Mastroeni and McCullough showed that the Chow ring of a matroid is Koszul [25]. In general, the graded Möbius algebra need not be Koszul, though LaClair, Mastroeni, McCullough, and Peeva proved that a graph G is strongly chordal if and only if the graded Möbius algebra of the graphic matroid $M(G)$ is Koszul [23]. In forthcoming work [17], McCullough and I explore the Koszul property for projective and affine geometries.

Theorem 1.1 ([17]). *If M is modular (e.g., a projective geometry), then $\text{GMA}(M)$ is Koszul.*

Proving Koszulness in this setting uses techniques from [23] to exhibit a quadratic Gröbner basis. For affine geometries, constructing a *Koszul filtration* seems to work, though a general filtration argument remains elusive. However, we can still show quadraticity:

Theorem 1.2 ([17]). *If M is an affine geometry of order $q > 2$, then $\text{GMA}(M)$ is quadratic.*

1.2. Betti Numbers. My collaborators and I have been working toward understanding the homological properties of these graded Möbius algebras. Using techniques from liaison theory, we have derived explicit formulas for all graded Betti numbers for the graded Möbius algebras of uniform matroids [12]. Since these algebras are combinatorially defined, we expect that the Betti numbers should have some combinatorial interpretation, too.

Project 1.3. *The graded Möbius algebra of a modular matroid is Gorenstein [17, 24], and thus is determined by the annihilator of the basis polynomial of the matroid [24]. Use the theory of inverse systems to compute the graded minimal free resolution of the graded Möbius algebra of a modular matroid.*

Generalizing projective and affine geometries, a *Sylvester* matroid is a matroid such that every pair of elements lies in a 3-circuit. We prove that if the graded Möbius algebra is quadratic and linearly presented, then the matroid is Sylvester [12]. We conjecture that if quadraticity is assumed, then the converse holds.

Project 1.4. *Let M be a simple matroid with at least 3 elements such that $\text{GMA}(M)$ is quadratic. Prove that if M is Sylvester, then $\text{GMA}(M)$ is linearly presented. What can be said about the Sylvester property and its relation (if any) to the Koszul property?*

A standard graded $\mathbb{K}$ -algebra A is *Artinian* if it is a finite-dimensional graded vector space over $\mathbb{K}$, i.e., $A_d = 0$ for $d \gg 0$. When there exists a general linear form $\ell \in A_1$ such that the multiplication map $\times \ell^d : A_i \rightarrow A_{i+d}$ is full rank for all $d, i \geq 0$ —that is, it is either injective or surjective—we say A has the *strong Lefschetz property* (SLP). If this map is full rank in all degrees for $d = 1$, then we say A has the *weak Lefschetz property* (WLP).

The SLP may be viewed as an algebraic stand-in for the landmark hard Lefschetz theorem: for a smooth n -dimensional complex projective variety X , the cup product of the k -th power of the class of a hyperplane section yields an isomorphism between $H^{n-k}(X)$ and $H^{n+k}(X)$ [20]. A corollary of this is that multiplication by a general linear form is injective up to degree n and surjective in higher degrees, a technique that has been used with great success in geometric combinatorics [8, 21, 28].

2.1. Betti Tables. Schenck and I investigated how the geometry of a set of points $X \subset \mathbb{P}^n$ influences whether A_X has the WLP, where A_X is the general Artinian reduction of the coordinate ring of X . We show that when all but one point lie on a unique degree- d hypersurface, A_X fails to have the WLP [18]. Then we generalize this to the following condition on the Betti table of A_X .

Definition 2.1. A Betti table B has an (n, d) -Koszul tail if its upper-left block has the form

	0	1	2	3	...	$n-2$	$n-1$	n
0	1	0	0	0	...	0	0	0
1	0	0	0	0	...	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$d-1$	0	0	0	0	...	0	0	0
d	0	n	$\binom{n}{2}$	$\binom{n}{3}$	...	$\binom{n}{n-2}$	n	1

Note that this is only the top portion of the Betti table. If B has an (n, d) -Koszul tail and is the Betti table for an Artinian algebra $\mathbb{K}[x_1, \dots, x_n]/I$, then we say B has a *maximal* (n, d) -Koszul tail.

Theorem 2.2 ([18]). *An Artinian algebra $A = \mathbb{K}[x_1, \dots, x_n]/I$ whose Betti table over $\mathbb{K}[x_1, \dots, x_n]$ has a maximal (n, d) -Koszul tail does not have the WLP; the map $\times \ell : A_d \rightarrow A_{d+1}$ is not full rank for any $\ell \in A_1$.*

Project 2.3. *A maximal Koszul tail is not sufficient for the WLP, however, as [1, Theorem 1.6] shows. What are conditions on the Betti table of A determine if A has (or does not have) the WLP?*

2.2. Connected Sums. The connected sum $A_1 \#_T A_2$ of two local Gorenstein rings A_1 and A_2 over a local Cohen-Macaulay algebra T was introduced in [5], and its connections to Lefschetz properties were studied in [22]. This algebraic operation is motivated by the corresponding topological connected sum operation, as Artinian Gorenstein algebras serve as algebraic analogs of cohomology rings of complex, compact, connected, oriented manifolds. For instance, when M_1 and M_2 are manifolds (with the aforementioned assumptions) glued along a common submanifold N , then $A_1 \#_T A_2 \cong H^{2*}(M_1 \#_N M_2)$, where $A_i = H^{2*}(M_i, \mathbb{K})$ and $T \cong H^{2*}(N, \mathbb{K})$. My collaborators and I give finer information about these connected sums by explicitly computing the Betti numbers in terms of the factors in the connected sum, and we show that the connected sum of doubling—a construction from the theory of Gorenstein liaison—is itself a doubling [4], i.e., it is realized as the cokernel of a map from the doubling construction.

The classical Tamari lattice was originally defined by a relation on bracketing expressions [29], but it has been studied in relation to triangulations of polygons, binary trees, and Dyck paths, and has rich connections to Catalan numbers [27]. Generalizations of the Tamari lattice such as the r -Tamari lattice [6] and the ν -Tamari lattice [26] have appeared throughout the literature in an effort to compute invariants of spaces appearing in the study of diagonal harmonics [19].

Continuing in this direction, my collaborators and I introduced the *stable Tamari lattice* $\mathcal{ST}_n = (\mathbb{N}^n, \leq_T)$, which can be obtained as a union over certain downsets in the r -Tamari lattice [10]. Although the definition of the stable Tamari order $\leq_T$ is technical, in specific contexts we can present $\leq_T$ nicely: if $\lambda \in \mathbb{N}^n$ is a partition and $\mathbf{a} \in \mathbb{N}^n$, then $\mathbf{a} \leq_T \lambda$ if and only if $\lambda - \mathbf{a}$ is a partition. The stable Tamari lattice is described as *stable* since $\mathbf{a} \leq \mathbf{b}$ if and only if $\mathbf{a} + \mathbf{1} \leq_T \mathbf{b} + \mathbf{1}$, where $\mathbf{1} = (1, 1, \dots, 1)$ is the all-ones vector.

My collaborators and I have been working to better understand $\mathcal{ST}_n$ and the enumerative combinatorics of $\mathcal{ST}_n$ and its *downsets* (or *lower order ideals*) such as the cover relations, their Möbius functions, their sizes, etc. [10]. For $\mathbf{b} \in \mathcal{ST}_n$, the downset $\mathbf{b}_\downarrow$ is the set of $\mathbf{a} \in \mathbb{N}^n$ with $\mathbf{a} \leq_T \mathbf{b}$. As a running example, Figure 1 is the Hasse diagram for $\mathbf{b}_\downarrow$ with $\mathbf{b} = (1, 2, 3)$.

We determine the cover relations in $\mathcal{ST}_n$, and we also show that the downsets in $\mathcal{ST}_n$ have a recursive decomposition into smaller downsets whose elements have fewer components; that is, given $\mathbf{b} \in \mathbb{N}^n$, we can express $|\mathbf{b}_\downarrow|$ as a sum of $|\mathbf{b}_{1\downarrow}|, \dots, |\mathbf{b}_{n\downarrow}|$ where $\mathbf{b}_i \in \mathbb{N}^{n-1}$.

Determining the sizes of the downsets in general is quite hard, but we have the following conjecture:

Conjecture 3.1. *For any $n \geq 1$, $\min_{\mathbf{b} \in \mathbb{S}_n} \{|\mathbf{b}_\downarrow|\} = C_{n+1}$, where C_{n+1} is the $(n+1)$ -st Catalan number. Moreover, the minimizers of this function are the elements $\mathbf{b} \in \mathbb{N}^n$ with exactly one peak.*

Conjecture 3.1 suggests a strong connection to parking functions, much-studied objects in the algebraic combinatorics community. Thus, establishing an explicit combinatorial bijection to other Catalan objects is a natural goal. In forthcoming work, we establish a bijection between elements in $\mathbf{b}_\downarrow$ and lattice points in a polytope, and then show that unimodal permutations achieve this conjectured minimum [9].

To better understand the overall structure of $\mathbf{b}_\downarrow$, we define two generating functions

$$f_{\mathbf{b}}(t) = \sum_{\mathbf{a} \in \mathbf{b}_\downarrow} t^{\text{ht}(\mathbf{a}, \mathbf{b})} \quad \text{and} \quad F_{\mathbf{b}}(t) = \sum_{[\mathbf{p}, \mathbf{q}] \subseteq \mathbf{b}_\downarrow} t^{\text{ht}(\mathbf{p}, \mathbf{q})},$$

where $\text{ht}(\mathbf{p}, \mathbf{q})$ is the maximum length of a chain from $\mathbf{p}$ to $\mathbf{q}$ in $\mathbf{b}_\downarrow$. Continuing with the running example of $\mathbf{b} = (1, 2, 3)$, we get

$$f_{\mathbf{b}}(t) = t^6 + t^5 + 2t^4 + 3t^3 + 3t^2 + 3t + 1 \quad \text{and} \quad F_{\mathbf{b}}(t) = t^6 + 4t^5 + 8t^4 + 15t^3 + 21t^2 + 21t + 14.$$

Conjecture 3.2. *For any $\mathbf{b} \in \mathbb{N}^n$, the coefficients of $f_{\mathbf{b}}(t)$ and $F_{\mathbf{b}}(t)$ are unimodal.*

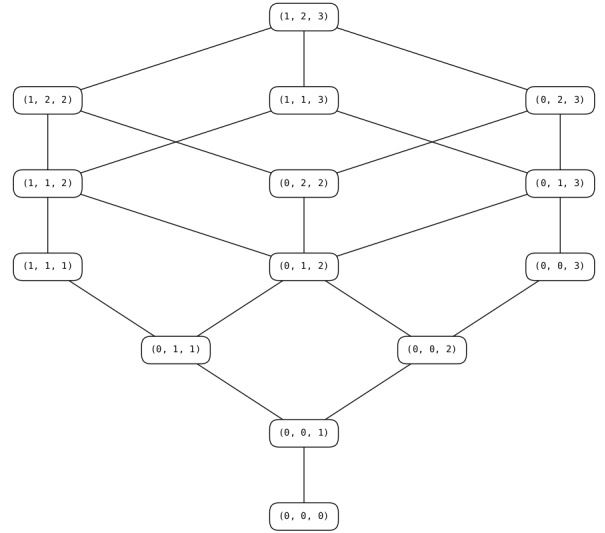


Figure 1: The Hasse diagram for $\mathbf{b}_\downarrow$ with $\mathbf{b} = (1, 2, 3)$.

4.1. Discipline-Based Education Research. Joint with Lanius and Gilroy, we developed the Course Analysis Alignment Tool (CAAT) to mathematically analyze the alignment of course objectives in classes and published our findings in [13]. Given a list of learning objectives, the CAAT has the instructor rank which objectives should be emphasized the most (ties are allowed), and then has them count how many times each objective appeared in an assignment, thus generating an observed ranking of objectives. The discrepancy or distance in these rankings is calculated and used to quantify how well an assignment aligns with the expectations of the instructor: the smaller the distance, the better the alignment (see Figure 2).

Alignment summaries are generated based on the quartiles of the distribution of distances. The first quartile (green) indicates an “Excellent” alignment, the second (yellow) an “Acceptable” alignment, and the last two (red) a “Poor” alignment. Each quartile is equally likely, so “Excellent” and “Acceptable” account for half of the possible scores and “Poor” the other half.

When analyzing the impact of the CAAT on the perception of uniform homework, we found that the CAAT affected participants’ definitions of homework quality. Especially with novice instructors, the CAAT shows promise as a professional development tool as it quantifies the alignment between proposed and assessed learning objectives.

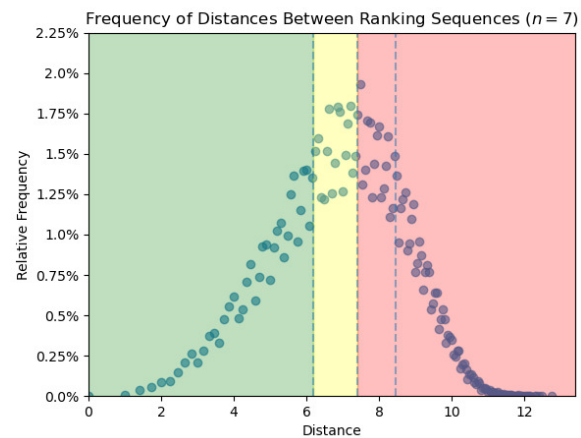


Figure 2: The distribution of distances on $n = 7$ learning objectives.

4.2. Contributions to Software. Much of my research is based on computations using software such as Python and the computer algebra system Macaulay2. Through a data-driven approach, I compute many examples to help formulate conjectures and build evidence in tackling research questions. For instance, when approaching the WLP project with Koszul tails, generating many examples of Betti tables is what allowed us to identify (1) that the Koszul tail was a key ingredient, and (2) that *maximal* Koszul tails were the crux of the matter.

I worked with a team on developing the `MatrixSchubert` package for Macaulay2 [3]. It features methods for constructing and analyzing matrix Schubert varieties; these include specialized methods to compute the regularity of a quotient formed by a Schubert determinantal (or alternating sign matrix) ideal, reduced pipe dreams associated to a permutation, and more. This package has been incorporated into version 1.23 of Macaulay2.

Realizing a need for a standard `Permutation` class in Macaulay2, I independently developed the `Permutations` package to formally incorporate them [15]; this package is included in version 1.24.11 of Macaulay2. I have also led a group in the development of the `YoungTableaux` package [16], with plans to merge it into the independently developed `Tableaux` package [14]. Later, I helped develop the `EquivariantEhrhart` package [11], with a target of being included in version 1.26.11 of Macaulay2.

I regularly contribute to core aspects of Macaulay2 and review pull requests on its official repository. Continuing to use, contribute to, and maintain open-source software to help enable data-driven discovery is central to my research methodology.

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